

[S4] Data Preperation I

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1. Berikan satu contoh kasus dimana menghapus outlier memperburuk keadaan
2. Apakah melakukan inputasi dengan mean / median dianggap netral? Jelaskan

nominal/kategorikal ada outlier

ordinal ada outlier

interval ada outlier

ratio ada outlier

data outlier hanya ada di sistem numerik

minority belum tentu outlier

outlier biasanya minority

kenapa outlier harus dihapus? agar datanya akurat

salah satu objektif dari ... adalah akurasi

klo datanya ga akurat → datanya tidak sama dengan realita

kanan outlier jangan di hapus? ketika outliernya merupakan data akurat atau data sebenarnya

ketika imputasi itu apakah mengimitasi atau representasi? harusnya representasi, tetap menjaga semantic dari distribusi datanya

namun sering kali ketika sintesa seperti dengan mean/median ini mengimitasi

variasi rendah → modelnya ga bisa untuk banyak kondisi

klo variasi tinggi → ...

klo variasi pas → modelnya bisa generalisasi

variasi itu untuk model mempelajari pola

klo tinggi: dia punya pola cuman di bagian tertentu

klo rendah: dia punya pola di semua bagian tapi sensitif di semuanya

klo pas: dia punya pola dan tidak terlalu sensitif

misal data1 nullnya 40% & variasinya kecil: sensitif di semua bagian

data2 ga ada null & variasi besar: sensitif di beberapa bagian

lebih baik pakai data1

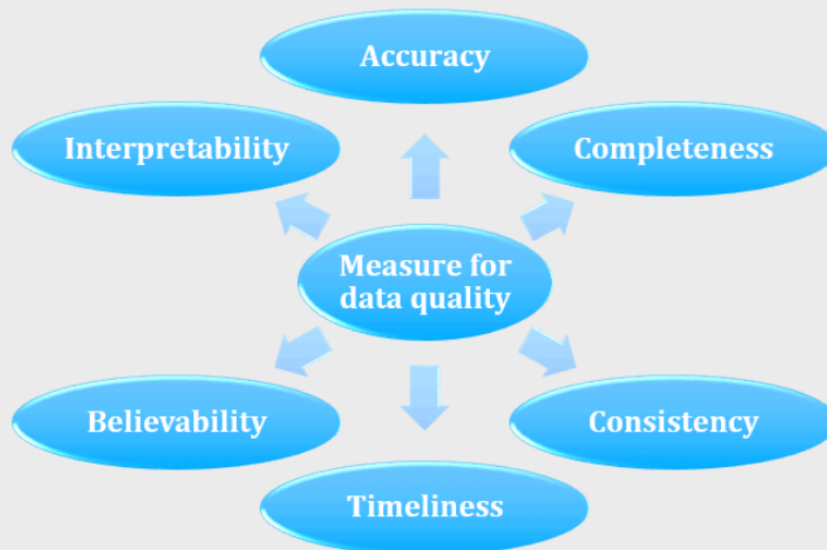


lah klo stress? nanti sering ke detect stress gimana di permodelannya tetep pake 40% null, tapi nanti di outputnya bisa disesuaikan agar persentase stressnya dikurangi (ya kayak di rubah lah outputnya) yang penting modelnya tetap akurat

ada kondisi dimana missing value, data yang existing masih bisa merepresentasi sebuah data,

makanya perlu memahami karakteristik dari kolom tersebut

DATA QUALITY ISSUES



ketika berbicara data? pertama yang di uji adalah data quality measure

baris → bagaimana encoding di ..., bisa dibaca, misal cwe cwo ditulisnya L P (ini contoh di baris, klo di kolum misalnya nama kolumnya jenis kelamin → jk. boleh aja asal dilengkapi dengan deskripsi nama kolum atau ada data directionary.

accuracy: $\text{delta}(\text{data}-\text{realita})$, selisih antara data dengan realita, semakin besar seslihnya → semakin tidak akurat, semakin kecil → akurat

bagaimana cara mengukur data dengan data asli? bukan data model

di plot aja, terus liat outliernya terus dibandingin dengan domain knowledge



apa itu domain knowledge? gimana cara bandinginnya

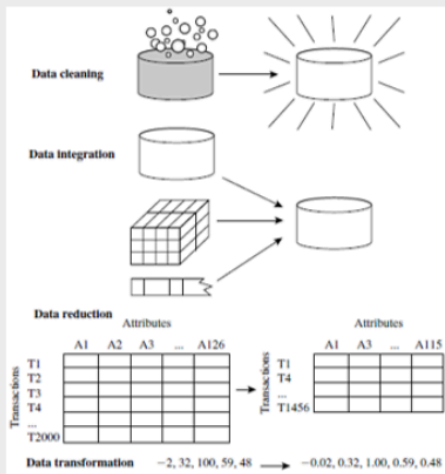
completeness: misalnya missing column, misal user ambil 3 matkul, tapi di datasetnya column matkulnya cuman 2, klo missing value bisa di input, klo missing column ga bisa, jangan hanya liat dimensi row, tapi bisa terjadi di column, bisa juga di dimensi tabel(?)

consistency: cowo cewe ga ada tengah tengah, bisa juga misal antar coilumn tidak bertentangan

timeliness: bisa pakai timestamp agar bisa tau datanya masih relevan dengan saat ini



MAJOR TASKS IN DATA PREPROCESSING



- **Data cleaning:** Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration:** Integration of multiple databases, data cubes, or files
- **Data reduction:** Dimensionality reduction, Numerosity reduction, Data compression
- **Data transformation:** Normalization, Concept hierarchy generation

DATA CLEANING

- Real world data is dirty: lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
 - Incomplete (missing values): lacking attribute values, lacking certain attributes of interest, or containing only aggregate data, e.g., Occupation=" "
 - Noisy: containing noise, errors, or outliers, e.g., Salary="-10" (an error)
 - Inconsistent: containing discrepancies in codes or names, e.g.,
 - Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - discrepancy between duplicate records
 - Intentional, e.g., disguised missing data

DATA CLEANING : MISSING VALUES

Incomplete (Missing) Values

- Data is not always available, e.g.: many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
 - Equipment malfunction
 - Inconsistent with other recorded data and thus deleted
 - Data not entered due to misunderstanding
 - Certain data may not be considered important at the time of entry
 - Not register history or changes of the data
- Missing data may need to be inferred

apa yang dicek ketika buang row?

misal fitur A B C

terus fitur A missing 5%

fitur B missing 5%

fitur C missing 5%

buang row yang missing di A & B & C

klo cuman missing di A dan B gimana?

kita anggap semua fitur penting

kapan di buang?

ketika kita yakin 2 hal:

1. tidak merubah distribusi data, misalnya di A memang tidak merusak distribusi data, tapi malah di B itu corenya jadi merusak distribusi datanya
2. sudah ada data lain yang mewakili distribusi datanya, misalnya klo kita hapus, masih ada data lain yang mewakilinya
misalnya
1 2 3
null 2 3
ini di hapus ada row null 2 3 nya karena udah diwakilin



DATA CLEANING : HANDLE MISSING VALUES

- Handle the missing data

Ignore the tuple

- usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably

Fill in the missing value

- manually
- automatically with a global constant: e.g., “unknown” or $-\infty$
- automatically with a measure of central tendency, e.g., mean or median
- automatically with the attribute mean or median for all samples belonging to the same class as the given tuple
- automatically with **the most probable value**: inference-based such as Bayesian formula or decision tree

klo ga bisa dibuang:

1. di ignore
2. di imputasi:
 - bisa pakai central tendency data (mean/median/modus): ketika tidak mengubah dispersi, karena bisa memporak porandakan model yang digunakan
 - klo ga bisa, pakai statistic_based



DATA CLEANING : NOISY DATA

- Noisy Data: A random error or variance in a measured variable
- Incorrect attribute values may be due to
 - Faulty data collection instruments
 - Data entry problems
 - Data transmission problems
 - Technology limitation
 - Inconsistency in naming convention
- Other data problems: duplicate records, incomplete, and inconsistent data

noisy mirip kayak null value, tapi noisy merupakan data yang tidak akurat misalnya akibat imputasi, data noisy yang tadinya di ujung jadi ke tengah jadi aman

ciri ciri distribusi yang aneh adalah distribusi yang aneh, ga keliatan pola gimana cara handlenya?

DATA CLEANING : HANDLE NOISY DATA

- Handle the noisy data

Binning

- First, sort the data and distribute into (equal-frequency) bins
- Then, one can **smooth by bin means, median, boundaries**, etc.

Regression

- Smooth by fitting the data into regression functions

Clustering

- Detect and remove outliers

Combined computer and human inspection

- Detect suspicious values and check by human (e.g., deal with possible outliers)

di binning:

DATA CLEANING : DISCRETIZATION BY BINNING

- Top-down split, unsupervised method
- **Equal-width** (distance) partitioning
 - Divides the range into N intervals of equal size: uniform grid
 - If A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A) / N$
 - The most straightforward, but outliers may dominate presentation
 - Skewed data is not handled well
- **Equal-depth** (frequency) partitioning
 - Divides the range into N intervals, each containing approximately same number of samples
 - Good data scaling
 - Managing categorical attributes can be tricky

data yang awalnya tinggi terus turun turun terus → cocok di binning

kapank pakai binning: ketika polanya tidak keliatan atau distribusinya ga homogen

- equal-width:

yang di samakan width nya

jarak x nya sama, y nya beda

equal depth ketika kita ingin menjaga kontingensi/keterlibatan sebuah nilai dalam proses permodelan

- equal-depth:

yang disamakan depth nya atau panjang batangnya

datanya normal

jarak y nya sama, x nya beda

panjang - panjang - panjang

comparable antar binning



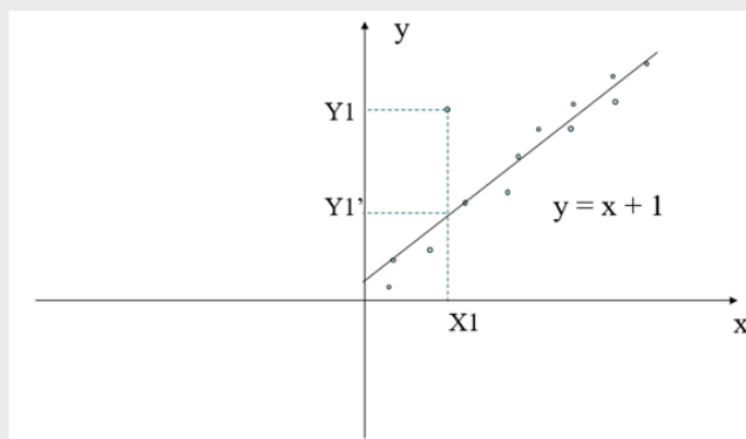
apakah binning dengan stratified mirip? apa bedanya? soalnya sama sama kayak membagi sesuai proporsinya

DATA CLEANING : EXAMPLE BINNING

- Example for data smoothing using binning method:
- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Method	Partition into Equal-depth bins	Smoothing by bin means	Smoothing by bin boundaries
Bin			
1	4, 8, 9, 15	9, 9, 9, 9	4, 4, 4, 15
2	21, 21, 24, 25	23, 23, 23, 23	21, 21, 25, 25
3	26, 28, 29, 34	29, 29, 29, 29	26, 26, 26, 34

DATA CLEANING : REGRESSION



imputasi juga bisa pakai regression

DATA CLEANING : AS A PROCESS

- Data Cleaning as a Process

Data discrepancy detection

- Use metadata (e.g., domain, range, dependency, distribution)
- Check field overloading
- Check uniqueness, consecutive, and null rule
- Use commercial tools:
 - Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

Data migration and integration

- Data migration tools: allow transformations to be specified
- ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface

- Integration of the two processes:
- Iterative and interactive (e.g., Potter's Wheels)

data cleaning: mau liat kualitas datanya seperti apa

integration:

DATA INTEGRATION

- Data integration

- Combines data from multiple sources into a coherent store

- Schema integration

- Integrate metadata from different sources, e.g., A.cust-id $\neq$ B.cust-#

- Entity identification problem:

- Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton

- Detecting and resolving data value conflicts

- For the same real world entity, attribute values from different sources are different
- Possible reasons: different representations, different scales, e.g., metric vs. British units

DATA INTEGRATION : REDUNDANCY

Data Redundancy

- Often occurs when integration of multiple databases
 - Object identification: the same attribute or object may have different names in different databases
 - Derivable data: one attribute may be a “derived” attribute in another table, e.g., annual revenue
- May be able to be detected by **correlation analysis** and **covariance analysis**
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

redudansi bisa di baris atau kolom

SUMMARY

- Measures for data quality: Accuracy, Completeness, Consistency, Timeliness, Believability, Interpretability
- Task in Data Preprocessing: Data cleaning, Data integration, Data reduction, Data transformation